Architecture Document

AI-Based Smart HVAC Controller

1. System Overview

The AI-Based Smart HVAC Controller is developed using MATLAB, Simulink, Stateflow, and Machine Learning. The architecture combines traditional HVAC control logic with AI-based mode prediction.

2. High-Level Architecture

User Inputs & Sensors
↓
Input Processing Layer
↓
Feature Processing Layer
↓
AI HVAC Predictor
↓
Stateflow HVAC Controller
↓
Actuator Command Layer
↓
Cabin Environment

3. Input Layer

Inputs:
• IGNITION_ON_OFF
• T_Cabin
• T_Ambient
• T_Setpoint
• Humidity_sensor
• Q_Solar
• HVAC_MODE_REQ

4. Input Processing Layer

Functions:
• Signal Validation
• Range Checking
• Temperature Error Calculation
• Humidity Processing
• Solar Load Processing

5. Feature Processing Layer

Generated Features:
• Temp_Error
• High_Humidity_Flag
• High_Solar_Flag
• HVAC_Enable

6. AI Prediction Layer

The AI controller uses: • T_Cabin • T_Setpoint • Humidity_sensor • Q_Solar • T_Ambient Output: • AI_HVAC_State State Mapping: 1 = IDLE 2 = COOLING 3 = HIGH_COOLING 4 = HEATING 5 =

DEFOG

7. Stateflow HVAC Controller

States: • OFF • IDLE • COOLING • HIGH_COOLING • HEATING • DEFOG The controller receives AI_HVAC_State and generates actuator commands.

8. Actuator Layer

Outputs: • Heater_Command • Compressor_Command • Blower_Speed • Recirc_Command • Airflow_Mode

9. Data Flow

Sensors → Input Processing → Feature Generation → AI Predictor → Stateflow Controller → Actuator Commands → HVAC System

10. Verification Strategy

• MIL Testing • Functional Test Cases • Baseline Testing • Simulink Test Manager Validation • AI Model Accuracy Verification

11. Future Architecture Enhancements

• Deep Learning HVAC Prediction • AUTOSAR Integration • SIL Testing • HIL Testing • Embedded Code Generation • Cloud-Based Analytics